

AMAZON SUPPLY CHAIN INTELLIGENCE & INVENTORY OPTIMIZATION PLATFORM

Analytics Report | Operations Analytics + Decision Support

Scope

Synthetic e-commerce supply-chain simulation covering validation, cleaning, SQL analytics, demand forecasting, inventory optimization and a 5-page Power BI control tower.

Data period

Sales simulation: 01 Jan 2025 - 31 Dec 2026

Latest inventory snapshot: 28 Dec 2026

Disclosure: Synthetic portfolio data - not Amazon internal data.

1. Executive Summary

The project was designed to demonstrate a complete analytics workflow: raw operational data -> data quality -> feature engineering -> SQL analysis -> demand forecasting -> inventory optimization -> Power BI decision support -> business recommendations. The centerpiece is the move from descriptive reporting to prescriptive actions at SKU x Warehouse level.

8.42M

Current inventory value

2.53%

Current stockout rate

1.93x

Average inventory turnover

202.78

Median days of inventory

3,801

Transfer opportunities

3.35MRecommended replenishment
value

Primary finding: the simulated network shows overstock and stockout risk at the same time. The strongest response is to rebalance existing inventory first, then use targeted replenishment for high-priority SKUs rather than increasing stock everywhere.

At-a-Glance Findings

Finding	Result / implication
Inventory imbalance	43,209 positions were classified for REDUCE INVENTORY, while 10,854 positions were Critical/High/Medium stockout risk.
Working-capital pressure	Median inventory coverage is about 203 days and 31.05% of positions contain stock with no recorded demand.
Rebalancing opportunity	3,801 SKU-warehouse positions were identified for TRANSFER STOCK before external replenishment.
Forecasting	Moving Average, Exponential Smoothing and Random Forest performed similarly; complexity did not create a material accuracy advantage.
Supplier / warehouse risk	High-risk suppliers and weaker warehouse delivery performance create avoidable service variability.

2. Data Foundation & Analytical Pipeline

The model uses eight interconnected synthetic tables designed to behave as one supply-chain system. The dataset includes product, supplier, warehouse, sales, inventory, procurement and forecast relationships.

Table	Business purpose	Rows
Products	SKU master, pricing, category, supplier, lead time	5,000
Suppliers	Supplier quality, reliability, capacity and lead time	300
Warehouses	Fulfillment locations, capacity and operating attributes	12
Inventory snapshots	Weekly SKU x warehouse inventory positions	3,120,000
Sales orders	Order-level fulfillment transactions	300,000
Sales order items	Product-level demand and returns	570,122
Purchase orders	Supplier replenishment activity	60,000
Demand forecasts	Historical forecast and actual demand records	149,986

Pipeline

Stage	Implementation
Validation	Row counts, nulls, duplicate IDs, foreign keys, invalid dates, negative quantities, price and inventory consistency.
Cleaning	Missing-value treatment, duplicate removal, date standardization, type conversion, outlier and invalid-record handling.
Feature engineering	Inventory efficiency, ABC class, product movement, margins, supplier risk and forecast error features.
SQL analytics	Inventory, supplier, warehouse, demand and optimization analysis in Microsoft SQL Server.
Forecasting	Moving Average, Exponential Smoothing and Random Forest on top 50 Product x Warehouse combinations.
Optimization	Safety stock, reorder point, EOQ, stockout / overstock risk, transfer and replenishment actions.
Visualization	Five-page Power BI control tower for executive and operational decision support.

3. Feature Engineering & Forecasting

Feature engineering converts transactional records into reusable business measures. ABC classification is based on cumulative net-sales contribution; fast/slow movers use sales-volume distribution; supplier risk combines reliability, on-time delivery, defects, late deliveries and quality.

Domain	Key engineered fields
Inventory	inventory_value, average_daily_demand, days_of_inventory, inventory_turnover, stockout_flag, overstock_flag, low_stock_flag, no_demand_stock_flag
Sales	gross_sales, net_sales, net_cost, gross_margin, net_units_sold, order_month, order_week, order_year
Products	ABC_class, fast_moving_flag, slow_moving_flag, high_value_flag
Suppliers	supplier_risk_score, late_delivery_flag, quality_risk_flag, actual_late_delivery_rate
Forecasts	forecast_error, MAPE, forecast_accuracy

Forecast Model Comparison

The forecasting layer deliberately compares simple, explainable models. The final 8 weeks are held out for time-based testing. Models are selected by error rather than by complexity.

Model	Average MAE	Average RMSE	Interpretation
Moving Average	8.39	10.29	Strong baseline; simple and transparent
Exponential Smoothing	8.39	10.34	Comparable performance with recency weighting
Random Forest	8.46	10.41	More complex, but no meaningful improvement

Business takeaway: use the simplest method that performs adequately, then improve accuracy through better demand segmentation, promotional variables, seasonality features and rolling validation rather than adding complexity for its own sake.

4. Inventory Optimization Engine

The optimization engine operates at SKU x Warehouse level and connects historical demand, future forecast demand, lead time, demand variability and current stock to inventory actions.

Metric	Business meaning
Planning daily demand	Higher of historical average daily demand and future forecast daily demand.
Demand variability	Weekly outbound-demand standard deviation used as uncertainty input.
Safety stock	Buffer stock scaled by demand variability, lead time and a 95% service-level assumption.
Reorder point	Planning daily demand x lead time + safety stock.
EOQ	Economic Order Quantity balancing ordering and annual holding cost.
Stockout risk	Critical / High / Medium / Low based on current stock vs lead-time demand and reorder point.
Overstock risk	High / Medium / Low based on no-demand inventory and days of inventory.
Recommended order quantity	Larger of inventory shortage to reorder point and EOQ when replenishment is required.

Model Assumptions

Assumption	Value
Approximate service level	95%
Service factor / Z-score	1.65
Ordering cost	50 per order
Annual holding rate	20% of unit cost

Optimization Action Distribution

Action	Positions
REDUCE INVENTORY	43,209
REORDER SOON	7,051
TRANSFER STOCK	3,801
HOLD	3,667
MONITOR	2,270
REORDER NOW	2

5. Power BI Supply Chain Control Tower

The dashboard is intentionally limited to five pages to keep each page focused and executive-friendly. The report combines current-state KPIs, trend analysis, root-cause views and direct inventory actions.

Page	Purpose	Screenshot file
1. Executive Supply Chain Overview	Inventory health, stockouts, turnover, supplier service, forecast accuracy and operational risk.	Executive Supply Chain Overview.png
2. Inventory Intelligence	ABC analysis, movers, inventory efficiency and SKU-warehouse decision matrix.	Inventory Intelligence.png
3. Supplier & Warehouse Performance	Supplier risk / lead time and warehouse fulfillment / delivery performance.	Supplier & Warehouse Performance.png
4. Demand & Forecast Intelligence	Actual vs forecast demand, seasonal trends, error and growing products.	Demand & Forecast Intelligence.png
5. Inventory Optimization Control Tower	Immediate-action SKUs, stockout / overstock risk, transfers and replenishment.	Inventory Optimization Control Tower.png

The dashboard should be interpreted as an operations decision screen, not only as a reporting artifact: Daily -> Critical / transfer risks; Weekly -> reorder and overstock review; Monthly -> supplier, warehouse, turnover and forecast performance.

6. Business Recommendations

Priority	Recommendation	Rationale
1	Reduce excess inventory	7.75M of current inventory value is associated with medium/high overstock-risk positions; working capital is the largest modeled opportunity.
2	Rebalance stock between warehouses	3,801 positions can potentially be served through transfers before external replenishment.
3	Prioritize A-class replenishment	Critical / high-risk high-value items should receive the strongest service protection.
4	Address supplier risk	10 suppliers have risk score ≥ 50 ; supplier variability should be reduced at source rather than masked with permanent excess stock.
5	Improve weak warehouse delivery performance	Phoenix and Queretaro show the lowest on-time delivery rates in the simulation, followed by Hyderabad and Los Angeles.
6	Plan seasonal peaks earlier	Demand rises around summer and year-end; temporary inventory buildup should be reversed after peak periods.
7	Improve forecast inputs	Simple models perform similarly; accuracy improvements should come from better features and segmentation.

30 / 60 / 90-Day Action Plan

Horizon	Actions
0-30 days	Review Critical/High stockout SKUs; execute valid transfers; stop severe overstock replenishment; focus on A-class products; review highest-risk suppliers.
30-60 days	Reduce C-class / slow-moving inventory; review no-demand stock; investigate weak warehouses; calibrate safety-stock / reorder assumptions; formalize performance scorecards.
60-90 days	Improve forecasting by segment; add promotional and seasonality variables; add transfer cost / lead time; improve SKU-level procurement linkage; build scenario analysis.

7. Assumptions, Limitations & Next Steps

This project is a synthetic portfolio simulation and is intended to demonstrate analytics thinking, data engineering discipline and decision-support design. It is not a production replenishment system.

Current limitation	Why it matters / improvement
No SKU-level product_id in purchase orders	Prevents direct SKU-to-PO replenishment reconciliation; add a purchase-order-line table in a future version.
No transfer cost or transfer lead time	Transfer recommendations identify opportunities but do not optimize network logistics cost.
No explicit lost-sales cost	Stockout exposure is an inventory-value proxy, not lost revenue or lost profit.
Synthetic ordering / holding assumptions	Reorder economics should be recalibrated if real business cost data becomes available.
Top-50 forecasting scope	Expand model coverage only after improving segmentation and time-series validation.

Conclusion

The strongest result of the project is the integrated decision chain: data quality -> operational KPIs -> demand forecasting -> inventory economics -> recommended actions. The model demonstrates that overstock and stockout risk can coexist in the same network, and that the better response is targeted rebalancing and prioritization rather than increasing inventory everywhere.

Portfolio narrative: "I built an end-to-end supply-chain analytics platform that validates millions of operational records, analyzes inventory / supplier / warehouse performance, compares demand forecasts, calculates safety stock, reorder points and EOQ, and converts the results into SKU-warehouse actions in a Power BI control tower."